SmartBrew AI - Task 2b Deliverable

Requirements and Planning

This document defines user stories, system requirements, and an initial product backlog for SmartBrew AI. The system is an intelligent coffee machine that supports personalized ordering, user profiles, AI recommendations, mood-based suggestions, inventory control, payments, receipts, and technician maintenance.

Custom Feature

The custom feature added by our group is **Loyalty Point Redemption**. It allows customers to redeem earned loyalty points during payment to reduce or cover the cost of a beverage.

User Stories

US-01 - Browse Menu and Nutrition

As a Customer, I want to browse beverage categories and inspect nutrition information so that I can choose a drink that fits my preferences and dietary needs.

US-02 - Log In

As a Customer, I want to log in using credentials or an NFC/card tap so that I can quickly access my personal preference profile.

US-03 - Create or Manage Profile

As a Customer, I want to create and manage my preference profile so that the machine can remember my favorite drink settings.

US-04 - Customize Beverage

As a Customer, I want to customize beverage parameters such as milk type, coffee strength, temperature, and syrup shots so that I receive the exact drink I want.

US-05 - Check Ingredient Availability

As a Customer, I want the machine to prevent unavailable drink selections so that I do not order a beverage that cannot be prepared.

US-06 - Generate Recommendations

As a Customer, I want to receive personalized beverage recommendations so that I can quickly choose a drink based on my preferences, history, time of day, and context.

US-07 - Analyze Mood

As a Customer, I want the machine to analyze my mood when recommendation mode is used so that it can suggest drinks that match my current state.

US-08 - Process Natural Language Order

As a Customer, I want to request a drink using free-form natural language so that I can order complex custom drinks without manually navigating every menu option.

US-09 - Process Payment

As a Customer, I want to pay using supported methods such as contactless card, digital wallet, or prepaid employee card so that I can complete checkout quickly.

US-10 - Redeem Loyalty Points

As a Customer, I want to redeem loyalty points earned from previous purchases so that I can receive discounts or free drinks.

US-11 - Receive Receipt

As a Customer, I want to receive a receipt by QR code or email so that I can keep proof of my purchase.

US-12 - Manage Inventory

As a Technician, I want to register ingredient refills and monitor waste levels so that the machine remains operational and does not offer drinks with depleted ingredients.

US-13 - View Error Logs

As a Technician, I want to view real-time machine error logs so that I can diagnose problems such as grinder jams, heater faults, and low ingredient thresholds.

System Requirements

ID	Name	Requirement
REQ-01	Browse Menu and Nutrition	The system shall display beverage categories, available menu items, and nutrition information through the touchscreen interface.
REQ-02	User Authentication	The system shall authenticate customers using credentials or NFC/card tap and load the associated preference profile after successful login.
REQ-03	Profile Management	The system shall allow customers to create and update preference profiles containing default drink settings such as milk type, strength, and temperature.
REQ-04	Beverage Customization	The system shall allow customers to customize beverage parameters, including milk type, coffee strength, temperature, syrup shots, and extra espresso shots.
REQ-05	Ingredient Availability	The system shall check ingredient levels against minimum thresholds and prevent selection of beverages whose required ingredients are depleted or too low.
REQ-06	Personalized Recommendations	The system shall generate personalized beverage recommendations using customer preferences, ordering history, time of day, weather context, and available menu items.
REQ-07	Mood Analysis	The system shall process facial expression input in memory to estimate the customer's mood and shall immediately discard facial images after inference.
REQ-08	Natural Language Ordering	The system shall use a local LLM module to translate free-form customer drink requests into explicit beverage customization parameters.
REQ-09	Payment Processing	The system shall support payment through contactless credit/debit cards, digital wallets, and prepaid employee cards.
REQ-10	Loyalty Point Redemption	The system shall allow logged-in customers to redeem loyalty points during payment to reduce or cover the beverage price.
REQ-11	Receipt Delivery	The system shall offer customers a receipt by QR code or email after a successful transaction.
REQ-12	Inventory and Waste Management	The system shall allow technicians in secure administrative mode to register ingredient refills, reset ingredient counters, monitor waste levels, and record waste disposal.
REQ-13	Error Logging and Diagnostics	The system shall display and store real-time machine error events, including grinder jams, heater faults, and low ingredient threshold warnings, for technician diagnostics.

Initial Product Backlog

ID	Name	Importance	Estimate	Description	How to Demo
PB-01	Browse Menu and Nutrition	High	3	Display beverage categories, drink details, and nutrition information on the touchscreen.	Open the menu, select a drink, and verify nutrition information is shown.
PB-02	User Authentication	High	3	Allow customers to log in using credentials or NFC/card tap and load their profile.	Tap a registered NFC card and verify the customer profile loads.
PB-03	Profile Management	Medium	3	Allow customers to create and update preference profiles.	Update default milk type and strength, save changes, and verify they are retained.
PB-04	Beverage Customization	High	5	Allow customers to modify drink parameters before ordering.	Select a drink, change milk type and syrup shots, and verify the recipe summary updates.
PB-05	Ingredient Availability Check	High	5	Prevent beverages from being selected when required ingredients are below the minimum threshold.	Set milk below threshold and verify milk-based drinks become unavailable.
PB-06	Personalized Recommendations	Medium	5	Generate recommendations based on preferences, order history, time, weather, and availability.	Log in as a customer and verify recommended drinks are ranked on the menu.
PB-07	Mood Analysis	Low	8	Analyze mood in memory and use the result for recommendations without storing facial images.	Trigger mood analysis and verify a mood-based recommendation appears while no image is stored.
PB-08	Natural Language Ordering	Low	8	Translate free-form drink requests into customization parameters using the local LLM module.	Enter a free-form request and verify suitable customization settings.
PB-09	Payment Processing	High	5	Support card, digital wallet, and prepaid employee card payments.	Complete an order using a contactless payment method and verify transaction success.
PB-10	Loyalty Point Redemption	Medium	5	Allow customers to redeem loyalty points during checkout.	Apply loyalty points during payment and verify the total price is reduced.
PB-11	Receipt Delivery	Medium	3	Provide receipts through QR code or email after payment.	Complete a payment and verify that a QR receipt is displayed or an email receipt is sent.
PB-12	Inventory and Waste Management	High	5	Allow technicians to register refills, reset counters, and monitor waste capacity.	Log in as technician, register a milk refill, and verify the milk counter resets.
PB-13	Error Logs and Diagnostics	Medium	5	Display and store error events for technician diagnostics.	Trigger a simulated grinder jam and verify that it appears in the error log.

Suggested First Sprint Focus

For an initial sprint, the team should focus on the highest-value foundation features:

1. PB-01 Browse Menu and Nutrition
2. PB-02 User Authentication
3. PB-04 Beverage Customization
4. PB-05 Ingredient Availability Check
5. PB-09 Payment Processing

These items form a basic usable ordering flow. AI recommendations, natural language ordering, loyalty redemption, and advanced diagnostics can be added after the core ordering and payment workflow is stable.